HOT SPOTS, THE FIRST NEUMANN EIGENSPACE, AND COMPLETE EXTREMAL SETS ON A REGULAR CUBE

Abstract. We give a self-contained proof of the hot spots property for the regular solid cube and determine the entire first positive Neumann eigenspace and the complete global extremal sets of each of its nonzero real elements. For $Q = (-1,1)^3$, the first positive eigenvalue is $\pi^2/4$ and has multiplicity three; its eigenspace has basis $\sin(\pi x_i/2)$, $1 \le i \le 3$. For $a \in \mathbb{R}^3 \setminus \{0\}$ and $u(x) = \sum_{i=1}^3 a_i \sin(\pi x_i/2)$, the maximum is $\sum_i |a_i|$ and is attained exactly where $x_i = \operatorname{sgn}(a_i)$ for every nonzero coefficient; coordinates with zero coefficient are free. The minimum is attained on the antipodal face. Thus the extremal sets are opposite closed faces, opposite closed edges, or opposite vertices, according as one, two, or three coefficients are nonzero. A complete product basis establishes spectral exhaustiveness, and an explicit quadratic deficit quantifies strictness away from the extremal sets. The figures illustrate all three possibilities.

Draft status. This candidate proof was obtained by our team on October 18, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

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1. Introduction and main result

Let

$$Q = (-1,1)^3 \subset \mathbb{R}^3. \tag{1.1}$$

We study real eigenfunctions of the Neumann problem

$$-\Delta u = \mu u \quad \text{in } Q, \qquad \partial_\nu u = 0 \quad \text{on } \partial Q. \tag{1.2}$$

The eigenvalue 0 corresponds to the constants. Throughout the paper, "first eigenfunction" means a nonzero eigenfunction for the *first positive* eigenvalue, denoted by μ_1. The hot spots property in the form considered here states that every global maximum and every global minimum of such an eigenfunction on $\overline{Q}$ lies on ∂Q.

Date: 18 October 2026.

2020 *Mathematics Subject Classification.* Primary 35P05; Secondary 35J05, 35B50.

Key words and phrases. Hot spots conjecture, Neumann Laplacian, regular cube, first positive eigenvalue, eigenspace multiplicity, extremal face.

The cube is an exactly solvable product-domain case. Its multiplicity still makes a precise formulation worthwhile: a statement about three selected eigenfunctions does not by itself describe every element of their span, and an extremal set need not consist of isolated points. The purpose of this paper is to include every coefficient pattern and every equality case in a single complete argument. For broader higher-dimensional hot spots results and the vector variational method on non-product domains, see Kennedy and Rohleder [1]. None of those general results is needed in the proof below.

Put

$$\phi_i(x) = \sin\left(\frac{\pi x_i}{2}\right), \qquad i = 1, 2, 3, \tag{1.3}$$

and, for $a = (a_1, a_2, a_3) \in \mathbb{R}^3$, set

$$u_a(x) = \sum_{i=1}^{3} a_i \phi_i(x), \qquad I(a) = \{i : a_i \neq 0\}, \qquad r(a) = \#I(a). \tag{1.4}$$

If $a \neq 0$, define two closed coordinate faces of $\overline{Q}$ by

$$\begin{aligned} F_+(a) &= \{x \in \overline{Q} : x_i = \operatorname{sgn}(a_i) \text{ for every } i \in I(a)\}, \\ F_-(a) &= \{x \in \overline{Q} : x_i = -\operatorname{sgn}(a_i) \text{ for every } i \in I(a)\}. \end{aligned} \tag{1.5}$$

Here a coordinate face is allowed to have dimension zero or one. Coordinates outside $I(a)$ range independently over $[-1, 1]$.

Theorem 1.1 (Complete classification on the cube)**.** *The first positive Neumann eigenvalue of Q is*

$$\mu_1 = \frac{\pi^2}{4}, \tag{1.6}$$

and its real eigenspace is exactly

$$E_1 = \operatorname{span}\{\phi_1, \phi_2, \phi_3\}. \tag{1.7}$$

In particular, $\dim E_1 = 3$*. For every $a \neq 0$, one has*

$$\max_{\overline{Q}} u_a = \sum_{i=1}^{3} |a_i|, \qquad \min_{\overline{Q}} u_a = -\sum_{i=1}^{3} |a_i|, \tag{1.8}$$

and the complete extremal sets are

$$\arg\max_{\overline{Q}} u_a = F_+(a), \qquad \arg\min_{\overline{Q}} u_a = F_-(a) = -F_+(a). \tag{1.9}$$

Both sets have affine dimension $3 - r(a)$ and lie in ∂Q. Moreover, ∇u_a vanishes nowhere in Q.

When $r(a) = 1$, the maxima fill one square face and the minima fill its opposite face. When $r(a) = 2$, they fill two opposite edges. When $r(a) = 3$, there is exactly one maximizing vertex and one minimizing vertex. Equality of two nonzero coefficient magnitudes does not create an additional free coordinate: the locations are determined by the signs and the zero set of the coefficient vector. In particular, $\phi_1 + \phi_2 + \phi_3$ has a unique maximum at $(1, 1, 1)$ and a unique minimum at $(-1, -1, -1)$.

Outline of the proof. Section 2 constructs a complete orthonormal product basis and uses weak testing to identify every Neumann eigenspace. It also proves the sharp mean-zero Poincaré inequality and its equality case, so the lower eigenvalue assertion is explicit. Section 3 computes the extremal values and all equality cases by one-variable monotonicity, completing the proof of Theorem 1.1. Section 4 records the face, edge, and vertex alternatives and illustrates them in color. Section 5 proves a quadratic separation estimate from the extremal faces. Section 6 transfers the classification to an arbitrarily positioned regular cube and gives the corresponding statement in any dimension.

2. A complete spectral description

2.1. **The weak formulation.** A weak Neumann eigenfunction with eigenvalue μ is a nonzero $u \in H^1(Q)$ satisfying

$$\int_Q \nabla u \cdot \nabla v \, \mathrm{d}x = \mu \int_Q uv \, \mathrm{d}x \qquad \text{for every } v \in H^1(Q). \tag{2.1}$$

Testing with u gives $\mu \int_Q u^2 \, \mathrm{d}x = \int_Q |\nabla u|^2 \, \mathrm{d}x \geq 0$; since $u \neq 0$, this implies $\mu \geq 0$. If $\mu = 0$, then $\nabla u = 0$ almost everywhere, so u is constant almost everywhere because Q is connected. If $\mu > 0$, testing with 1 gives $\int_Q u = 0$. This formulation encodes the Neumann condition weakly on all open faces and requires no choice of normal at edges or vertices. The functions constructed below are smooth on all of $\mathbb{R}^3$. Proposition 2.3 will show that every weak eigenfunction agrees almost everywhere with a finite sum of these functions. We use this smooth representative when discussing values and extrema on $\overline{Q}$.

2.2. **One-dimensional modes and product completeness.** On $(-1, 1)$ define

$$\chi_0(t) = 2^{-1/2}, \qquad \chi_k(t) = \cos\left(\frac{k\pi(t+1)}{2}\right) \quad (k \geq 1), \qquad \lambda_k = \frac{k^2\pi^2}{4} \quad (k \geq 0). \tag{2.2}$$

Lemma 2.1 (The interval basis)**.** *The functions $\{\chi_k : k \in \mathbb{N}_0\}$ form an orthonormal basis of $L^2(-1, 1)$ and satisfy*

$$-\chi_k'' = \lambda_k \chi_k, \qquad \chi_k'(-1) = \chi_k'(1) = 0. \tag{2.3}$$

Proof. For $j, k \geq 1$, the cosine product identity after the change of variables $s = (t + 1)/2$ gives $\int_{-1}^1 \chi_j(t)\chi_k(t) \, \mathrm{d}t = \delta_{jk}$. Also, $\int_{-1}^1 \chi_k = 0$ for $k \geq 1$, while $\int_{-1}^1 \chi_0^2 = 1$. This proves orthonormality. Differentiation proves (2.3).

For completeness, change variables to $s = (t + 1)/2 \in [0, 1]$. The map $s \mapsto \cos(\pi s)$ is a homeomorphism from $[0, 1]$ to $[-1, 1]$. Thus every continuous function f on $[0, 1]$ has the form $f(s) = g(\cos(\pi s))$ with g continuous on $[-1, 1]$. By polynomial approximation, g is a uniform limit of polynomials. To express those polynomials in cosine modes, set $T_0(z) = 1$, $T_1(z) = z$, and $T_{k+1}(z) = 2zT_k(z) - T_{k-1}(z)$ for $k \geq 1$. The cosine addition identity gives $T_k(\cos\theta) = \cos(k\theta)$. Since T_k has degree k with nonzero leading coefficient, $T_0, \ldots, T_m$ span all polynomials of degree at most m. Thus every polynomial in $\cos(\pi s)$ is a finite linear combination of $\cos(k\pi s)$, $k \geq 0$. Consequently their linear span is dense in $C([0, 1])$ in the uniform norm, hence dense in $L^2(0, 1)$. Changing variables back proves completeness. □

For $k = (k_1, k_2, k_3) \in \mathbb{N}_0^3$, set

$$X_k(x) = \chi_{k_1}(x_1)\chi_{k_2}(x_2)\chi_{k_3}(x_3), \qquad \Lambda_k = \frac{\pi^2}{4}(k_1^2 + k_2^2 + k_3^2). \tag{2.4}$$

Lemma 2.2 (The product basis)**.** *The family $\{X_k : k \in \mathbb{N}_0^3\}$ is an orthonormal basis of $L^2(Q)$. Each X_k satisfies the weak Neumann eigenvalue equation with eigenvalue Λ_k.*

Proof. Fubini's theorem and Lemma 2.1 give orthonormality. For completeness, finite linear combinations of rectangle indicators are dense in $L^2(Q)$: a continuous function on $\overline{Q}$ can be approximated uniformly by functions constant on a sufficiently fine rectangular grid, and continuous functions are dense in $L^2(Q)$. Each rectangle indicator is a product $f_1(x_1)f_2(x_2)f_3(x_3)$ with $f_i \in L^2(-1, 1)$. By Lemma 2.1, choose finite interval-mode sums $p_{i,m}$ converging to f_i in $L^2(-1, 1)$. Fubini's theorem gives the product norm identity $\| \prod_i h_i(x_i)\|_{L^2(Q)} = \prod_i \|h_i\|_{L^2(-1,1)}$. Applying it to the three terms in the telescoping difference $\prod_i p_{i,m} - \prod_i f_i$ shows that the difference tends to zero in $L^2(Q)$; here each sequence $\|p_{i,m}\|_2$ is bounded because it converges in L^2. Every product $\prod_i p_{i,m}$ is a finite linear combination of the X_k, proving completeness.

Direct differentiation gives $-\Delta X_k = \Lambda_k X_k$. On a face $x_i = \pm 1$, its normal derivative vanishes by (2.3). Integration by parts on the box gives the weak identity for smooth test functions. Density of smooth restrictions in $H^1(Q)$ extends it to every H^1 test function: such approximations are obtained by reflecting across the faces to extend an $H^1(Q)$ function to a neighborhood of $\overline{Q}$, and then mollifying. Both sides of the weak identity are continuous in the $H^1(Q)$ norm. □

Proposition 2.3 (All eigenspaces)**.** *The Neumann eigenvalues on Q are exactly the numbers*

$$\frac{\pi^2}{4}(k_1^2 + k_2^2 + k_3^2), \qquad (k_1, k_2, k_3) \in \mathbb{N}_0^3, \tag{2.5}$$

with multiplicities given by the number of triples producing the same number. More precisely, for each eigenvalue μ its eigenspace is

$$E_\mu = \operatorname{span}\{X_k : \Lambda_k = \mu\}. \tag{2.6}$$

Proof. Let u satisfy (2.1) and write $c_k = \int_Q u X_k \, \mathrm{d}x$. Testing the equation for u with X_k, and the equation for X_k with u, gives

$$(\mu - \Lambda_k) c_k = 0 \qquad (k \in \mathbb{N}_0^3). \tag{2.7}$$

At least one coefficient is nonzero because the product family is complete and $u \neq 0$. Thus $\mu = \Lambda_k$ for some k. Equation (2.7) makes all coefficients with $\Lambda_k \neq \mu$ vanish. Only finitely many triples have $\Lambda_k = \mu$, since $0 \leq k_i \leq 2\sqrt{\mu}/\pi$ for every such triple. Completeness therefore gives

$$u = \sum_{\Lambda_k = \mu} c_k X_k \quad \text{in } L^2(Q) \text{ and almost everywhere in } Q.$$

This finite sum supplies a smooth representative of u on $\overline{Q}$. It is the unique continuous representative there, since continuous functions agreeing almost everywhere in Q agree throughout $\overline{Q}$. The converse in (2.6) follows from Lemma 2.2 and linearity. Finally, the product modes are orthonormal, so the indicated number of triples is the dimension of the eigenspace. □

The least positive integer of the form $k_1^2 + k_2^2 + k_3^2$ is 1. It is attained exactly at $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$. Since

$$\chi_1(t) = -\sin(\pi t/2), \qquad X_{(1,0,0)} = -\tfrac{1}{2}\phi_1, \quad X_{(0,1,0)} = -\tfrac{1}{2}\phi_2, \quad X_{(0,0,1)} = -\tfrac{1}{2}\phi_3, \tag{2.8}$$

Proposition 2.3 proves (1.6) and (1.7). In particular, this identifies every first eigenfunction, not only a selected symmetry basis.

2.3. **The sharp Poincaré inequality and equality.** The established spectral modes also control the Dirichlet energy of every mean-zero $H^1(Q)$ function, with a precise equality case.

Proposition 2.4 (Sharp Poincaré inequality)**.** *For every $v \in H^1(Q)$ with $\int_Q v = 0$,*

$$\int_Q |\nabla v|^2 \, \mathrm{d}x \geq \frac{\pi^2}{4} \int_Q |v|^2 \, \mathrm{d}x. \tag{2.9}$$

Equality holds if and only if $v \in E_1$, including $v = 0$.

Proof. Let $c_k = \int_Q v X_k \, \mathrm{d}x$ and $S_N = \{0, \ldots, N\}^3$ for $N \in \mathbb{N}_0$. Set $P_N = \sum_{k \in S_N} c_k X_k$. Testing the weak equations from Lemma 2.2 with v and with P_N, and using orthonormality, gives

$$\int_Q \nabla v \cdot \nabla P_N \, \mathrm{d}x = \int_Q |\nabla P_N|^2 \, \mathrm{d}x = \sum_{k \in S_N} \Lambda_k |c_k|^2.$$

Consequently,

$$0 \leq \int_Q |\nabla (v - P_N)|^2 \, \mathrm{d}x = \int_Q |\nabla v|^2 \, \mathrm{d}x - \sum_{k \in S_N} \Lambda_k |c_k|^2.$$

The partial sums $\sum_{k\in S_N}\Lambda_k|c_k|^2$ are nonnegative and increase as $N\to\infty$, so

$$\int_Q |\nabla v|^2\,\mathrm{d}x \geq \sum_{k\in\mathbb{N}_0^3}\Lambda_k|c_k|^2. \tag{2.10}$$

The mean-zero assumption gives $c_{(0,0,0)}=0$. For every other triple, $\Lambda_k\geq\pi^2/4$. Parseval's identity for the complete product basis now proves (2.9).

If equality holds in (2.9), the nonnegative sum

$$0\leq\sum_{k\neq(0,0,0)}\left(\Lambda_k-\frac{\pi^2}{4}\right)|c_k|^2\leq\int_Q|\nabla v|^2\,\mathrm{d}x-\frac{\pi^2}{4}\int_Q|v|^2\,\mathrm{d}x=0$$

vanishes by (2.10) and Parseval's identity. Therefore $c_k=0$ except at the three triples in (2.8), so $v\in E_1$. Conversely, every $v\in E_1$ satisfies equality by its weak eigenvalue equation. □

Remark 2.5 (Canonical coefficients). By (2.8) and Lemma 2.2, $\phi_i/2$ is an orthonormal basis of E_1. Consequently,

$$\int_Q\phi_i\phi_j\,\mathrm{d}x=4\delta_{ij},\qquad \|u_a\|^2_{L^2(Q)}=4\sum_i a_i^2,\qquad a_i=\frac{1}{4}\int_Q u_a\phi_i\,\mathrm{d}x. \tag{2.11}$$

If e_i denotes the ith coordinate unit vector, then e_i is a face center of this cube and $u_a(e_i)=a_i$. At a vertex $\varepsilon\in\{-1,1\}^3$, one has $u_a(\varepsilon)=\sum_i a_i\varepsilon_i$.

3. Exact global extrema of every first eigenfunction

Lemma 3.1 (The one-dimensional equality cases)**.** *The function $s(t)=\sin(\pi t/2)$ is a strictly increasing homeomorphism from $[-1,1]$ onto $[-1,1]$. For every $t\in[-1,1]$ and $b\neq0$,*

$$b\,s(t)\leq|b|,\quad \textit{with equality exactly when } t=\operatorname{sgn}(b), \tag{3.1}$$

and

$$b\,s(t)\geq-|b|,\quad \textit{with equality exactly when } t=-\operatorname{sgn}(b). \tag{3.2}$$

For $b=0$, both equalities hold for every $t\in[-1,1]$.

Proof. The derivative $s'(t)=(\pi/2)\cos(\pi t/2)$ is strictly positive on $(-1,1)$, so the mean value theorem gives strict monotonicity even when one or both comparison points are endpoints. Together with continuity and $s(-1)=-1$, $s(1)=1$, this makes s a continuous bijection of compact intervals, hence a homeomorphism. Since $|s(t)|\leq1$, one has $b\,s(t)\leq|b|$. For $b\neq0$, equality is equivalent to $s(t)=\operatorname{sgn}(b)$, whose unique preimage is $t=\operatorname{sgn}(b)$. Replacing b by $-b$ gives the lower bound and its equality condition. For $b=0$, both sides are identically zero. □

Proposition 3.2 (Complete equality classification)**.** *For every $a\in\mathbb{R}^3\setminus\{0\}$, the extremal values and sets are given by (1.8) and (1.9).*

Proof. For every $x\in\overline{Q}$, Lemma 3.1 gives

$$-\sum_i|a_i|\leq u_a(x)\leq\sum_i|a_i|. \tag{3.3}$$

To identify the upper equality set, write

$$\sum_i|a_i|-u_a(x)=\sum_i\bigl(|a_i|-a_i s(x_i)\bigr). \tag{3.4}$$

Every summand is nonnegative, and those with $a_i=0$ vanish identically. The sum is zero if and only if each summand with $a_i\neq0$ is zero, which by (3.1) is equivalent to $x\in F_+(a)$. This set is nonempty: choose $x_i=\operatorname{sgn}(a_i)$ for $i\in I(a)$ and choose the remaining coordinates arbitrarily in $[-1,1]$. Thus the upper bound is attained exactly on $F_+(a)$, proving both the maximum and its full equality set.

Since $x \mapsto -x$ is a bijection of $\overline{Q}$ and $u_a(-x) = -u_a(x)$, it maps the complete maximizing set onto the complete minimizing set. Hence the minimum is $-\sum_i |a_i|$ and its equality set is $-F_+(a) = F_-(a)$. □

Completion of the proof of Theorem 1.1. The eigenvalue, multiplicity, and eigenspace were established in Section 2, and Proposition 3.2 gives the global extrema. Exactly $r(a)$ independent coordinates are fixed in either extremal set, so its affine dimension is $3 - r(a)$. Because $a \neq 0$, at least one coordinate is fixed at 1 or -1, and both sets belong to ∂Q.

Finally,

$$\partial_i u_a(x) = \frac{\pi}{2} a_i \cos\left(\frac{\pi x_i}{2}\right). \tag{3.5}$$

For $x \in Q$, each cosine in (3.5) is strictly positive, so the right side is nonzero whenever $i \in I(a)$. Since $I(a)$ is nonempty, $\nabla u_a(x) \neq 0$ everywhere in Q. Fermat's necessary condition for a differentiable function therefore excludes every interior local maximum or minimum. □

Corollary 3.3 (The supremum norm of the first eigenspace)**.** *The coefficient map $a \mapsto u_a$ is a linear isometry from $(\mathbb{R}^3, \|\cdot\|_1)$ onto $(E_1, \|\cdot\|_{L^\infty(Q)})$:*

$$\|u_a\|_{L^\infty(Q)} = \sum_i |a_i|. \tag{3.6}$$

Proof. The map is a linear bijection by (1.7). For $a \neq 0$, put $M = \max_{\overline{Q}} |u_a| = \sum_i |a_i|$ by (1.8). Clearly the essential supremum on Q is at most M. Choose $x_* \in \overline{Q}$ attaining M. For each $\varepsilon > 0$, continuity gives $\delta > 0$ such that $|u_a| > M - \varepsilon$ throughout $B_\delta(x_*) \cap Q$. Because $x_* \in \overline{Q}$, this intersection is a nonempty open subset of $\mathbb{R}^3$; it contains a ball and therefore has positive measure. Thus the essential supremum is at least $M - \varepsilon$; letting $\varepsilon \downarrow 0$ proves the norm identity. For $a = 0$ it holds trivially. □

4. Faces, edges, and vertices: the geometric classification

The alternatives established in Theorem 1.1 are collected in Table 1. "Closed face" and "closed edge" include their boundaries. There is no excluded genericity assumption on a.

Nonzero coefficients	Maximizing set	Minimizing set
$r(a) = 1$	One closed square face	The opposite square face
$r(a) = 2$	One closed edge	The opposite edge
$r(a) = 3$	One vertex	The opposite vertex

Table 1. The complete extremal-set classification for $u_a \neq 0$.

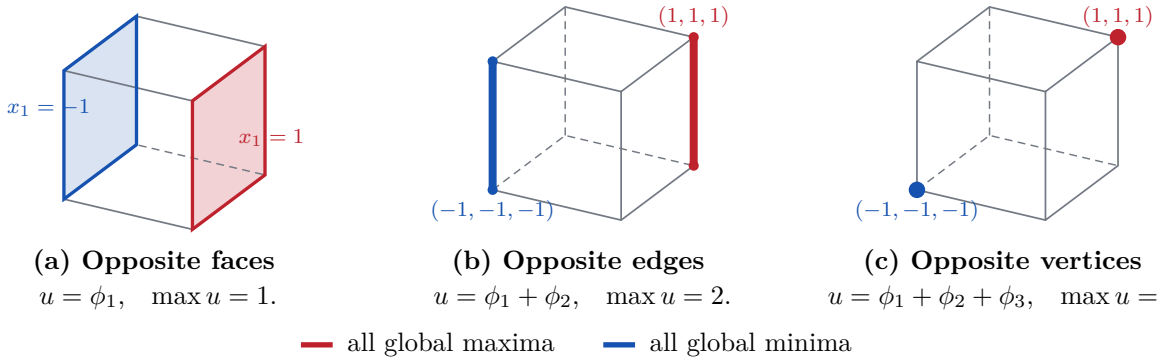
Figure 1 illustrates the three alternatives using positive nonzero coefficients. If S is a signed permutation matrix, representing coordinate reflections and permutations, then $SQ = Q$ and the oddness of sine gives

$$u_a(Sx) = u_{S^\intercal a}(x).$$

The transformed function belongs to E_1 by (1.7), so these operations preserve the first Neumann eigenvalue equation. Changing signs and permuting the coordinates produces every placement of the listed extremal sets within the fixed cube.

For example, changing the sign of the second coefficient in panel (b) gives

$$\arg\max(\phi_1 - \phi_2) = \{(1, -1, t) : -1 \le t \le 1\}, \qquad \arg\min(\phi_1 - \phi_2) = \{(-1, 1, t) : -1 \le t \le 1\}.$$



(a) Opposite faces $u = \phi_1, \quad \max u = 1.$

(b) Opposite edges $u = \phi_1 + \phi_2, \quad \max u = 2.$

(c) Opposite vertices $u = \phi_1 + \phi_2 + \phi_3, \quad \max u = 3.$

FIGURE 1. Complete extremal sets, with maxima in red and minima in blue. In (a), the colored regions are the entire faces $x_1 = \pm 1$. In (b), the red edge is $\{(1, 1, t) : -1 \le t \le 1\}$ and the blue edge is $\{(-1, -1, t) : -1 \le t \le 1\}$. In (c), only the two marked vertices are extremal. The projections and vertex coordinates are the same in all panels; dashed gray segments are hidden edges.

Similarly, $2\phi_1 - \phi_2 + 5\phi_3$ has its unique maximum at $(1, -1, 1)$ and its unique minimum at $(-1, 1, -1)$. The coefficient signs and zero set determine the extremal-set geometry. For any fixed such pattern, the nonzero magnitudes determine the extremal values and the profile of the eigenfunction.

5. A QUANTITATIVE DEFICIT FROM THE EXTREMAL FACES

The explicit formula also quantifies strictness away from the maximizing and minimizing sets, including arbitrarily close to a face, an edge, or a vertex. No boundary regularity theorem at a corner is required. Throughout this section, dist denotes Euclidean distance.

Proposition 5.1 (Quadratic separation)**.** *Fix $a \neq 0$, write $M = \sum_i |a_i|$, and put*

$$m_a = \min_{i \in I(a)} |a_i| > 0, \qquad A_a = \max_{i \in I(a)} |a_i|.$$

For every $x \in \overline{Q}$,

$$\frac{m_a}{2} \operatorname{dist}(x, F_+(a))^2 \le M - u_a(x) \le \frac{\pi^2 A_a}{8} \operatorname{dist}(x, F_+(a))^2. \tag{5.1}$$

The analogous inequalities for the minimum are

$$\frac{m_a}{2} \operatorname{dist}(x, F_-(a))^2 \le M + u_a(x) \le \frac{\pi^2 A_a}{8} \operatorname{dist}(x, F_-(a))^2. \tag{5.2}$$

Proof. For $i \in I(a)$, write $\varepsilon_i = \operatorname{sgn}(a_i)$ and

$$t_i = 1 - \varepsilon_i x_i \in [0, 2].$$

Then

$$a_i \sin(\pi x_i/2) = |a_i| \cos(\pi t_i/2),$$

so the exact identity

$$M - u_a(x) = \sum_{i \in I(a)} |a_i| \left(1 - \cos \frac{\pi t_i}{2}\right) \tag{5.3}$$

holds. On $[0, \pi/2]$, concavity places $\sin s$ above the chord joining its endpoint values, so $\sin s \ge 2s/\pi$. Also, $\sin s = \int_0^s \cos z \, dz \le s$. Substituting $s = \pi t/4$ for $0 \le t \le 2$ gives

$$\frac{t^2}{2} \le 1 - \cos \frac{\pi t}{2} = 2 \sin^2 \frac{\pi t}{4} \le \frac{\pi^2 t^2}{8}. \tag{5.4}$$

For every $y \in F_+(a)$, the active coordinates contribute $\sum_{i\in I(a)} |x_i-\varepsilon_i|^2$ to $|x-y|^2$, and the remaining coordinates contribute a nonnegative sum. This lower bound is attained by taking $y_i = \varepsilon_i$ for $i \in I(a)$ and $y_j = x_j$ for $j \notin I(a)$; the latter choices are allowed because $x \in \overline{Q}$. Consequently,

$$\operatorname{dist}(x, F_+(a))^2 = \sum_{i\in I(a)} |x_i - \varepsilon_i|^2 = \sum_{i\in I(a)} t_i^2. \tag{5.5}$$

Multiplying (5.4) by $|a_i|$ and summing in (5.3), then bounding each active weight below by m_a and above by A_a, gives (5.1) by (5.5). Replacing a by $-a$ leaves M, m_a, A_a unchanged, replaces u_a by $-u_a$, and changes $F_+(a)$ to $F_-(a)$, yielding (5.2). □

In particular, the deficit is strictly positive at every point outside the appropriate extremal face. Directions corresponding to zero coefficients remain exactly flat, as they must: those directions parametrize the entire extremal set. With $t_i = 1 - \operatorname{sgn}(a_i)x_i$ for $i \in I(a)$, the exact identity (5.3) also gives the local expansion

$$M - u_a(x) = \frac{\pi^2}{8} \sum_{i\in I(a)} |a_i| t_i^2 + O\left(\sum_{i\in I(a)} |a_i| t_i^4 \right)$$

as the active coordinate distances tend to zero. Indeed, Taylor's theorem at 0 through degree three, with the fourth derivative of $t \mapsto \cos(\pi t/2)$ bounded in absolute value by $(\pi/2)^4$, gives

$$\left| 1 - \cos\frac{\pi t}{2} - \frac{\pi^2 t^2}{8} \right| \le \frac{\pi^4 t^4}{384}, \qquad 0 \le t \le 2,$$

so the remainder above is bounded in absolute value by $\frac{\pi^4}{384} \sum_{i\in I(a)} |a_i| t_i^4$.

6. Arbitrary position, scale, and dimension

Corollary 6.1 (Every regular solid cube)**.** *Let*

$$\Omega = x_0 + \ell R Q, \qquad x_0 \in \mathbb{R}^3, \quad \ell > 0, \quad R \in O(3).$$

Then the side length is 2ℓ, the first positive Neumann eigenvalue is $\pi^2/(4\ell^2)$, and its eigenspace has basis

$$\phi_i^\Omega(x) = \sin\left(\frac{\pi}{2\ell} \big(R^{\mathsf{T}}(x - x_0)\big)_i \right), \qquad i = 1, 2, 3. \tag{6.1}$$

For $a \in \mathbb{R}^3 \setminus \{0\}$ and $u = \sum_i a_i \phi_i^\Omega$, the maximizing and minimizing sets are $x_0 + \ell R F_+(a)$ and $x_0 + \ell R F_-(a)$, respectively, and the extremal values remain $\pm \sum_i |a_i|$.

Proof. Let $T(y) = x_0 + \ell R y$, whose inverse is $T^{-1}(x) = \ell^{-1} R^{\mathsf{T}}(x - x_0)$. Composition with this invertible affine map gives a linear bijection between $H^1(\Omega)$ and $H^1(Q)$. For $u, v \in H^1(\Omega)$, put $w = u \circ T$ and $z = v \circ T$. The weak chain rule gives $\nabla_y w(y) = \ell R^{\mathsf{T}} \nabla_x u(T(y))$ almost everywhere, and similarly for z. Orthogonality of R gives $|\det DT| = \ell^3$, including when $\det R = -1$. Hence

$$\int_\Omega \nabla u \cdot \nabla v \, \mathrm{d}x = \ell \int_Q \nabla w \cdot \nabla z \, \mathrm{d}y, \qquad \int_\Omega uv \, \mathrm{d}x = \ell^3 \int_Q wz \, \mathrm{d}y.$$

Since the test functions correspond bijectively, u has Neumann eigenvalue μ on Ω if and only if w has eigenvalue $\ell^2 \mu$ on Q. This correspondence preserves eigenspace dimensions and the ordering of eigenvalues because $\ell^2 > 0$. Theorem 1.1 therefore gives the asserted first positive eigenvalue and basis, with $\phi_i^\Omega = \phi_i \circ T^{-1}$. Moreover, T maps $\overline{Q}$ bijectively onto $\overline{\Omega}$, and $u \circ T = u_a$ for the stated linear combination. Thus it maps the complete maximizing and minimizing sets of u_a onto those of u, preserving their values, which proves the remaining claims. □

Corollary 6.2 (The n-dimensional cube)**.** *Let $n \geq 1$ be an integer. For $Q_n = (-1,1)^n$, the first positive Neumann eigenvalue is $\pi^2/4$ and has multiplicity n. Every real first eigenfunction has the unique form*

$$u(x) = \sum_{i=1}^{n} a_i \sin(\pi x_i/2).$$

For $a \neq 0$, its maximum and minimum are $\pm \sum_i |a_i|$, attained exactly on the two antipodal coordinate faces obtained by fixing $x_i = \pm \operatorname{sgn}(a_i)$ for each nonzero coefficient. Those faces have dimension $n - \#\{i : a_i \neq 0\}$.

Proof. Use the n-fold product of the interval basis in Lemma 2.1: for $k \in \mathbb{N}_0^n$, let $X_k^{(n)}(x) = \prod_{i=1}^n \chi_{k_i}(x_i)$ and $\Lambda_k^{(n)} = \frac{\pi^2}{4} \sum_{i=1}^n k_i^2$. The proof of Lemma 2.2 applies with n factors: these modes form an orthonormal basis of $L^2(Q_n)$ and satisfy the weak Neumann equation with eigenvalue $\Lambda_k^{(n)}$ for every test function in $H^1(Q_n)$.

For a nonzero weak eigenfunction u with eigenvalue μ, write $c_k = \int_{Q_n} u X_k^{(n)} \, \mathrm{d}x$. Testing the equations for u and $X_k^{(n)}$ against each other, as in Proposition 2.3, gives

$$(\mu - \Lambda_k^{(n)}) c_k = 0 \qquad (k \in \mathbb{N}_0^n).$$

Completeness ensures that some $c_k \neq 0$, so $\mu = \Lambda_k^{(n)}$ for some k. The least positive such value is $\pi^2/4$, attained exactly at the n coordinate unit vectors e_i; their modes are themselves eigenfunctions. At this eigenvalue, the coefficient identity and completeness give the finite expansion

$$u = \sum_{i=1}^{n} c_{e_i} X_{e_i}^{(n)}, \qquad X_{e_i}^{(n)} = -2^{-(n-1)/2} \sin(\pi x_i/2).$$

The equality holds in $L^2(Q_n)$ and almost everywhere. As in Proposition 2.3, this finite sum supplies the unique continuous representative on $\overline{Q_n}$ used for extrema. Orthonormality of the n modes gives both the multiplicity and the uniqueness of the displayed sine coefficients.

The proof of Proposition 3.2 uses only the one-dimensional equality conditions of Lemma 3.1 and a finite sum of nonnegative deficits, so it applies to these n summands without change. It gives the stated extrema and their complete equality sets. In either set, exactly $r = \#\{i : a_i \neq 0\}$ coordinates are fixed, while the remaining $n - r$ range independently over $[-1, 1]$, giving affine dimension $n - r$. This also covers $n = 1$, when the two extremal sets are the two endpoints. □

References

[1] J. B. Kennedy and J. Rohleder, *On the hot spots conjecture in higher dimensions*, arXiv:2410.00816v2, 2025. https://arxiv.org/abs/2410.00816v2.